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RESEARCH PAPER

Influences of the process parameter and thermal cycles on the quality of 308L stainless steel walls produced by additive manufacturing utilizing an arc welding source

Van Thao Le[1] · Dinh Si Mai[1] · Manh Cuong Bui[2] · Kilian Wasmer[3] · Van Anh Nguyen[4] · Duc Manh Dinh[1] · Van Canh Nguyen[5] · Duong Vu[6]

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Abstract

In this paper, the effects of the deposition speed and thermal cycles in gas-metal arc-welding (GMAW) additive manufacturing on the quality of as-built 308L stainless steel thin walls were investigated. The results exhibit that the deposition speed and thermal cycles play a crucial role in the quality of produced parts. An increase in deposition speed results in an improvement in the surface waviness. The surface waviness (S_a) decreases from 286 to 138 µm as the deposition speed increases from 0.2 to 0.4 m/min. On the other hand, the growth of microstructures in the walls fabricated with different deposition speeds shows a similar trend. The microstructure of as-built 308L-stainless-steel walls consists of dominant columnar/equiaxed dendrites of austenite and small amount of ferrite remaining in grain boundaries. The deposition speed mainly influences the grain size in microstructures. In the middle part of the walls, an augmentation in the deposition speed leads to a decrease in the secondary dendrite arm spacing, which results in an enhancement in mechanical properties of the walls. The microhardness and ultimate tensile strength increase from 153 ± 7.16 to 164 ± 8.96 HV0.1 and from 483 ± 4.24 to 518 ± 2.83 MPa, respectively, when the deposition speed increases from 0.2 to 0.4 m/min.

Keywords Additive manufacturing · Arc welding · 308L stainless steel · Deposition speed · Thermal cycle · Microstructure · Tensile property

1 Introduction

Because of the capacity of producing complex and customized components, additive manufacturing (AM) is becoming a key technology of Industry 4.0 [1, 2]. AM can significantly mitigate wastes as compared with conventional machining processes, particularly in the case of machining aerospace components with a high buy-to-fly ratio [3, 4]. Currently, powder-bed fusion (PBF) and powder-based directed-energy deposition (P-DED) processes are the most popular commercial metallic AM technologies. However, due to their limited building volume, PBF-based AM systems are normally adopted for products with small to medium sizes and customized and complex shapes [5]. Meanwhile, P-DED seems beneficial to larger parts with medium complexity. This technique, however, reveals a low level of material utilization efficiency [6]. The price of metallic powder used in PBF and P-DED processes is also very expensive, and only few kinds of metallic alloys are available [7, 8]. Hence, the

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✉ Van Thao Le
thaomta@gmail.com

[1] Advanced Technology Center, Le Quy Don Technical University, Hanoi, Vietnam

[2] Faculty of Mechanical Engineering, Le Quy Don Technical University, Hanoi, Vietnam

[3] Laboratory for Advanced Materials Processing (LAMP), Swiss Federal Laboratories for Materials Science and Technology (Empa), CH-3602 Thun, Switzerland

[4] Welding Engineering and Laser Processing Centre, Cranfield University, Cranfield, UK

[5] Hanoi University of Industry, 298 Cau Dien, Bac Tu Liem, Hanoi, Vietnam

[6] Mechanical Engineering Faculty, Duy Tan University, Da Nang, Vietnam

production of large-scale components by PBF and P-DED is still a practical challenge.

Recently, wire-plus-arc additive manufacturing (WAAM) has emerged as an effective technique to manufacture large-dimensional components [9, 10]. Among the energy sources used in WAAM, gas-metal-arc welding (GMAW) is the most preferable for producing metal parts with large dimensions, because it features a highly deposited rate, 2–10 kg h^{-1} [7, 11, 12]. In this process, the electrical arc is generated between the wire tip and the substrate (Fig. 1a). The metal wire is melted by the high heat of the electrical arc, and the molten metal is deposited into the workpiece in the X–Y plane to create layers. The part is built up along the *Z* direction by adding the material according to the deposition paths. Within the deposition process, the shielding gas is used to isolate the arc and the deposited metal from the surrounding ambience.

As an emerging AM technology, the control of WAAM process conditions to obtain expected part qualities is still a major challenge. The part quality strongly depends on the processing parameters, including wire-feed speed, arc voltage, welding current, travel speed/deposition speed, heat input, and deposition strategies [10, 11]. Many research works have been conducted to explore the effects of such factors on the external and internal properties of components [8, 13]. For instance, Dinovitzer et al. [14] found that the deposition speed and welding current mainly influenced the shape of weld beads and the microstructural characteristics of as-built nickel alloy. An increase in deposition speed or a decrease in welding current led to a mitigation in the depth of melting through and an augmentation in surface roughness. Xiong et al. [15] also examined the impact of wire feed speeds and deposition speeds on the characteristics of surfaces of low-carbon steel walls. They observed that with an augmentation in wire-feed speed the surface smooth decreased. Meanwhile, the increase in deposition speed enables enhancing the surface smooth.

Rodrigues et al. [16] indicated that the heat input significantly influenced the surface waviness and microstructural and mechanical characteristics of HSLA steel parts. An increase in heat input could reduce the surface roughness. This is because of the fine wettability of the current layer on the previous one. On the other hand, there was no noticeable change in microstructures between the samples built with different heat input levels. The heat input also had a significant effect on the geometry of low-carbon steel walls, but this factor did not much influence the formation and growth of microstructures [17]. However, in the case of WAAM austenitic stainless steel [18], increasing in heat input also led to an increase in primary dendrites' spacing. The weak corrosion performance was also observed for the samples produced with high heat input levels [18].

Su et al. [19] produced aluminum-magnesium thick walls with different wire feed speeds and deposition speeds. They showed that the grain morphology in the interlayer and within the layer could be modified by controlling wire feed speed and deposition speed values. Moreover, the tensile strengths exhibited an augmentation trend when *v* increased. Yinghui et al. [20] also conducted a study to analyze the impact of travel speed on the microstructural and mechanical characteristics of an aluminum alloy. The main findings of their study showed that the deposition speed had an

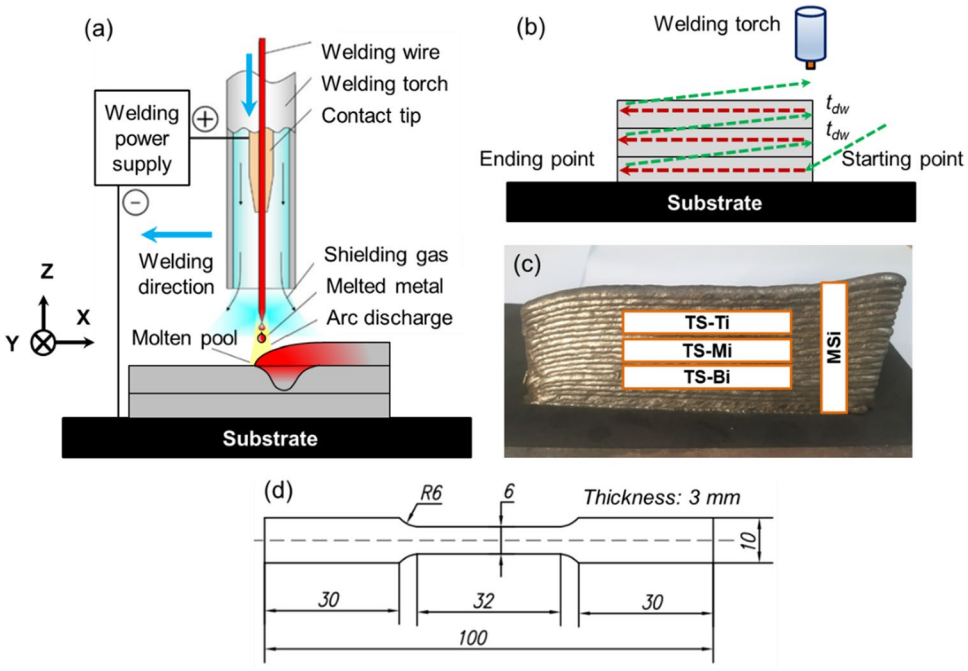


Fig. 1 **a** Illustration of the WAAM process using GMAW [23], **b** deposition paths for building the wall samples, **c** positions for cutting the metallurgical specimen (MSi) and tensile testing specimens (TS-Ti, TS-Mi, TS-Bi), and **d** dimensions of tensile specimens

opposite effect on the size and the volumetric proportion of equiaxed grains. The precipitation of Al_2Cu phases could be promoted at high levels of the deposition speed. In terms of tensile properties, the samples built with high deposition speed levels feature higher tensile strengths. Recently, Tawfik et al. [21] also examined the effect of deposition speed on the properties of aluminum-magnesium alloy (ER5356) fabricated by GTAW-AM. These authors also found that fine and equiaxed grains were formed in the samples fabricated with high deposition speed levels, which results in high tensile strengths of the samples. This is due to the cooling and solidification rates of the welding pool which were faster when the travel speed increased. However, the variation in the deposition speed caused a small effect on the microhardness. When the travel speed increased from 200 to 500 mm/min, the average microhardness of the fabricated samples increased from 70.71 ± 4.61 to 73.01 ± 3.65 HV, with only a small difference of 2.3 HV.

In this paper, the influences of the process parameter and thermal cycles in GMAW-based AM on the quality of stainless steel (308L) walls are discussed. The 308L stainless steel is chosen, because this steel is commonly utilized in shipbuilding, petroleum, and mining fields due to its excellent corrosion resistance and mechanical characteristics. The goal is to enrich the knowledge on the WAAM of stainless steels, and to give some guidelines for process planners and operators to select a proper deposition speed for a specified application. For this aim, three single-pass-multilayer walls have been fabricated with three levels of the deposition speed, while other process parameters (i.e., arc voltage and welding current) were kept constant. Subsequently, the geometry and microstructural and mechanical properties of the walls have been evaluated through the geometrical assessment, the microstructure observation, and the mechanical property analysis. The thermal cycles in the walls were analyzed via simulations based on the finite element method.

2 Materials and methods

2.1 Materials and the fabrication of thin wall samples

In this study, three hot-rolled 316L stainless steel plates with a size of 200 mm × 100 mm × 10 mm (length × width × height) were utilized as the substrates for the build of thin wall samples. A stainless-steel wire (GM-308L) supplied by Kim Tin Corporation was utilized as the consumable electrode. The GM-308L wire has the main chemical composition as follows: 0.06% (max) of C, 0.65% (max) of Si, 0.03% (max) of P and S, (9.00~11.00)% of Ni, (19.50~21.00)% of Cr, 0.75% (max) of Cu, 0.50% (max) of Mo, (1.00~2.50)% of Mn, and balance of Fe (in %wt.) [22].

A welding robot (Panasonic TA1400) was used for the GMAW-based AM process. A power source (Panasonic YD-350GR3) was used to produce the electrical arc between the part and the wire (Fig. 1a). In this system, the adjustment of the wire feed speed is performed in an automatic manner based on the welding current. The wire-feed speed usually increases as the welding current increases [15]. The thin-wall samples of twenty layers have been constructed according to the single direction method, as described in Fig. 1b. The processing parameters utilized for fabricating thin walls are presented in Table 1.

The length of welding paths was programed to be 120 mm. The deposition speed is varied to generate different heat input levels in the build of the thin-wall samples. Other parameters were set with the same value for building all three samples.

2.2 Characterization methods

After the GMAW-based AM process, contamination and oxidization on the wall surfaces were cleaned. The geometry of all three walls was achieved by a Range 7 3D laser scanner (Konica Minolta). The wall width, the wall height, and the layer height of the walls were measured directly from the scanned data by using Geomagic Design X software (Fig. 2). The morphological characteristics of wall surfaces were evaluated by utilizing OmniSurf3D software (as shown in Fig. 3).

Thereafter, the specimens for the observation of metallurgical and tensile properties were cut from the thin-wall samples (Fig. 1c). The specimens (MSi) were used for analyzing microstructures, microhardness, and chemical composition. Before the microstructure observation, the surfaces corresponding to the wall cross section of these specimens were ground, polished, and chemically etched with the solution mixed by 20 ml of HCl, 10 ml of HNO_3, and 30 ml of water within 10 s. The images of microstructures were captured with an optic microscope (Axio A2M, Carl Zeiss). Additionally, the secondary dendrite arms' spacing (SDAS) in microstructures was measured from the microstructure images by using the line intercept method [24, 25] and

Table 1 The parameters used to produce the thin-wall samples

Parameters	Value and unit
Arc voltage (U_a)	20 V
Welding current (I_c)	120 A
Deposition speed (v)	0.2; 0.3; and 0.4 m/min
Shielding gas (99.99% argon) flow rate	15 L/min
Distance from the torch tip to the part	12 mm
The dwell time applied between two adjacent layers	40 s

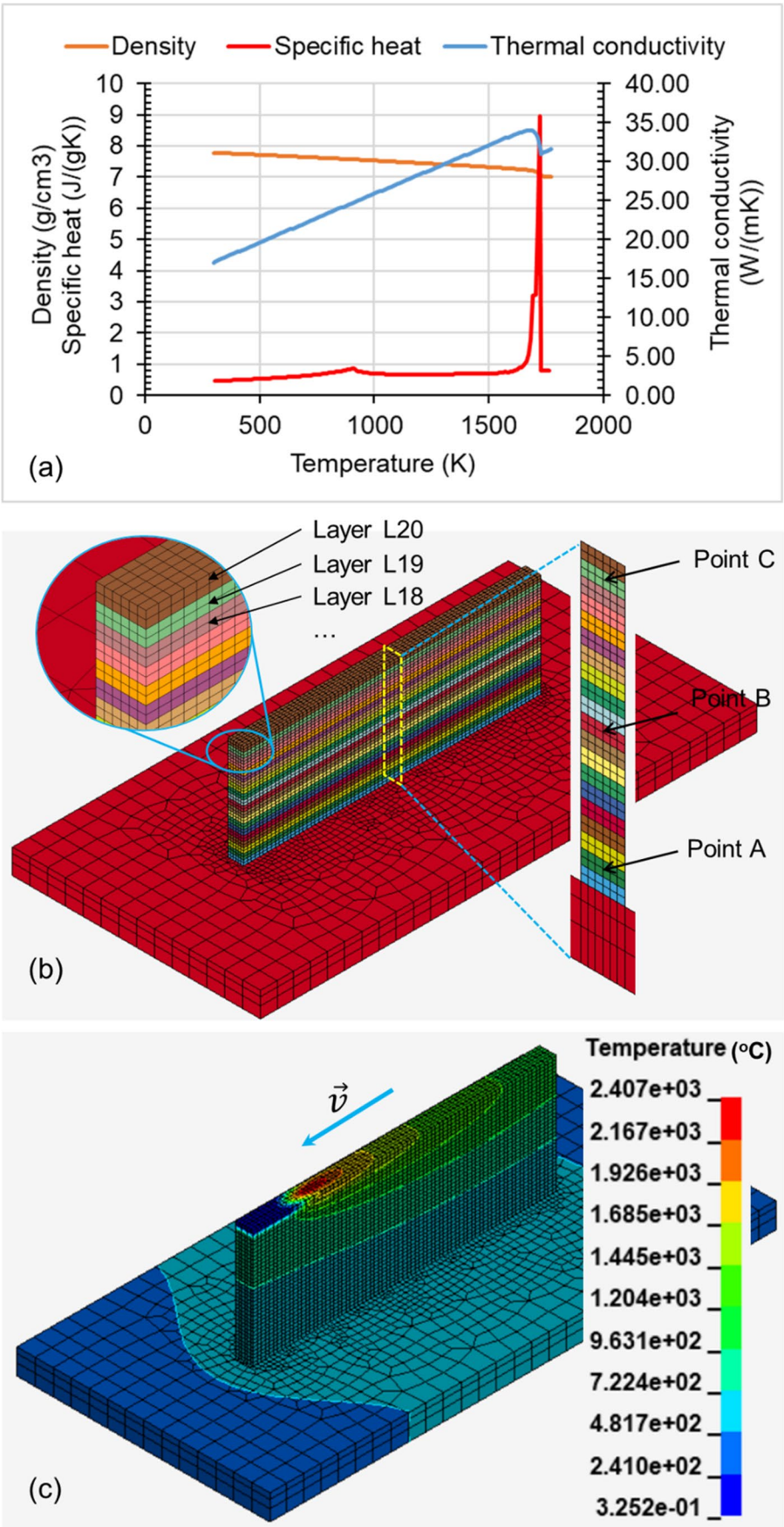


Fig. 2 **a** Thermophysical properties of SS 308L. **b** The mesh model used for the simulation and the points for observing the thermal cycle curves during the WAAM process, and **c** the simulated temperature evolution results

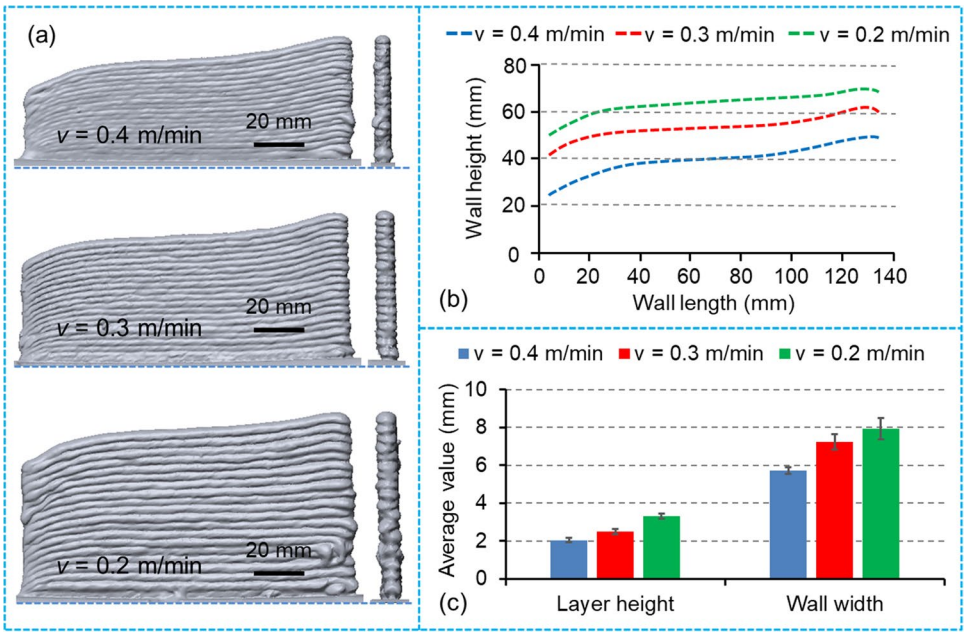


Fig. 3 **a** The shape of the walls produced with $v = 0.4$, 0.3 and 0.2 m/min, **b** the comparison of the wall height, and **c** the layer height and the wall width between the walls

ImageJ software. According to this method, SDAS = l/N, where l presents the line length (μm), and N depicts the number of secondary-dendrite arms. The microhardness (HV0.1) has been tested by a Vickers hardness machine (Duramin-2 model, Struers). Each indentation was performed with 980.7 mN in load and 10 s in dwell time. Lastly, the MSi specimens were used to explore the chemical composition by utilizing a ARL3460OES spectrometer of Thermo Scientific.

The tensile specimens (TS-Ti, TS-Mi, and TS-Bi) were designed with respect to the ASTM standard (E8/E8M-13) (Fig. 1d) [26]. The tensile tests have been conducted by using a tensile tester (Model-5967B12620, INSTRON) at room temperature. The crosshead displacement velocity was set at 10 mm/min. A scanning electron microscope machine (SU3500) was utilized to capture the morphologies of fractured surfaces of the tensile specimens.

2.3 FE-based simulation method for predicting the thermal evolution

This study aims to analyze the thermal evolution and thermal distribution in the thin-wall samples. Finite element (FE) simulations of the WAAM process have been conducted by using LS-DYNA software (Fig. 2b, c). The technique of element deactivation/activation was applied in the simulation. Namely, all elements of deposited layers are deactivated before the simulation as they do not exist before the depositing process. The deactivated elements are activated sequentially following the movement of the welding torch with the deposition speed given in Table 1. The double ellipsoidal power density distribution model of Goldak [27] was adopted to model the WAAM heat source. Among various models of heat source, this Goldak model enables capturing steeper gradients of the temperature in front of the arc compared to those at the edge of the melting pool. This model is widely and efficiently applied in previous studies [28–30]. Accordingly, the heat flux for the points in the first semi-ellipsoid lied in front of the weld arc is expressed by Eq. (1). On the other hand, the heat flux for the points in the second semi-ellipsoid covered the rear section of the arc is depicted by Eq. (2).

$$q_f(x,y,z) = \frac{6\sqrt{3}f_f Q}{a_f bc\pi\sqrt{\pi}}\exp\left(-\frac{3x^2}{a_f^2} - \frac{3y^2}{b^2} - \frac{3z^2}{c^2}\right) \quad (1)$$

$$q_r(x,y,z) = \frac{6\sqrt{3}f_r Q}{a_r bc\pi\sqrt{\pi}}\exp\left(-\frac{3x^2}{a_r^2} - \frac{3y^2}{b^2} - \frac{3z^2}{c^2}\right) \quad (2)$$

where a_f, a_r, b and c are geometric parameters; f_r and f_f are the heat input proportions in the rear and front parts, $f_r + f_f = 2$. The values of these parameters reported in [31] were adopted. On the other hand, the heat $Q = \eta U_a I_c$ is computed from U_a, I_c, and η given in Table 1.

The thermal physical properties of 308L stainless steel, including thermal conductivity, density, and specific heat used for FE simulations, were calculated based on its chemical composition (Table 1) with the help of JMatPro software (as shown in Fig. 2a). In WAAM, the interchange of heat takes place between built walls, the substrate, and environmental surroundings. As a result, the convection and radiation can be modeled by using the coefficients of convection and emissivity, and the Stefan-Boltzmann constant. The emissivity and convection coefficients were fixed to 0.9 and

30 W/(m^2.K), respectively, for the top surface of the substrate and deposited layers, while the convection coefficient was set at 300 W/(m^2.K) for the other substrate surfaces. The Stefan-Boltzmann constant of 5.67×10^{-8} W/(m^2 K) was adopted [31].

The 3D models of the walls composed of 20 layers were constructed based on the geometries of the substrates and the walls. The layer height and width for each wall were taken from the measurement results (Sect. 3.1). Linear-brick elements with eight nodes were utilized for the mesh generation. A fine mesh was used for the deposited layers and the surrounding areas of the substrate that is close to the wall. The meshes became coarser in the directions far away from the wall (Fig. 2b). The interesting points (A, B, and C) for extracting the thermal evolution cycles are in the middle layers L2, L10, and L18, and they distribute in the central line of the transversal section, as illustrated in Fig. 2b.

3 Results and discussion

3.1 Shape observation

Figure 3a shows the shapes of the thin-wall samples fabricated with three deposition speed levels ($v = 0.4$, 0.3, and 0.2 m/min). For all samples, the wall height tends to decrease gradually from the beginning position to the ending position of the deposition path (Fig. 3a and b). This is a common result of deposits with the single direction deposition method. The difference in height of the single weld bead was not compensated by such a deposition method [32, 33]. To achieve a regular height of the walls, a zigzag direction deposition method is chosen [33, 34] or adaptive control systems of deposited height need to be developed [10, 35].

As shown in Fig. 3b, the wall height reduces as the travel speed increases. Similarly, it is also revealed that the average width of the walls and the average layer height increase with a decrease in the travel speed (Fig. 3c). The thin-wall samples built with $v = 0.4$, 0.3, and 0.2 m/min have an average width value of 5.73 ± 0.18, 7.23 ± 0.41, and 7.93 ± 0.57 mm, respectively, and an average layer height of 2.04 ± 0.11, 2.50 ± 0.12, and 3.32 ± 0.14 mm, respectively. As the deposition speed increases, the heat input and the volume of deposited materials per length unit decrease, leading to a reduction in the width, the height, and the layer height of the walls [16, 36].

Moreover, it is noticed that the width of the walls is the narrowest at the bottom (i.e., the first layer). As the number of deposited layers increases, the wall width becomes more regular. This is due to the fast cooling to the cold substrate within the deposition of the first layer. For the next layers, the temperature of the substrate augments and the heat accumulation occurs, thus preventing the tightening of the molten metal and improving the wettability of deposited layers. This phenomenon is commonly observed in the WAAM process of single-pass multi-deposited-layer walls [16, 20, 37, 38].

Figure 4 shows the morphology characterization of the walls' side surfaces. It is revealed that the wall constructed with high travel speed has a lower surface waviness.

The wall built with $v = 0.2$ m/min features the highest values of all the area roughness parameters ($S_a = 286$ µm, $S_q = 347$ µm, and $S_z = 1985$ µm) of the side surfaces, followed by the walls built with $v = 0.3$ ($S_a = 175$ µm, $S_q = 220$ µm, and $S_z = 1405$ µm) and $v = 0.4$ m/min ($S_a = 138$ µm, $S_q = 173$ µm, and $S_z = 1003$ µm), respectively (Fig. 4). The obtained results are good in agreement with those reported in previously published works [15, 34, 39]. As the deposition speed

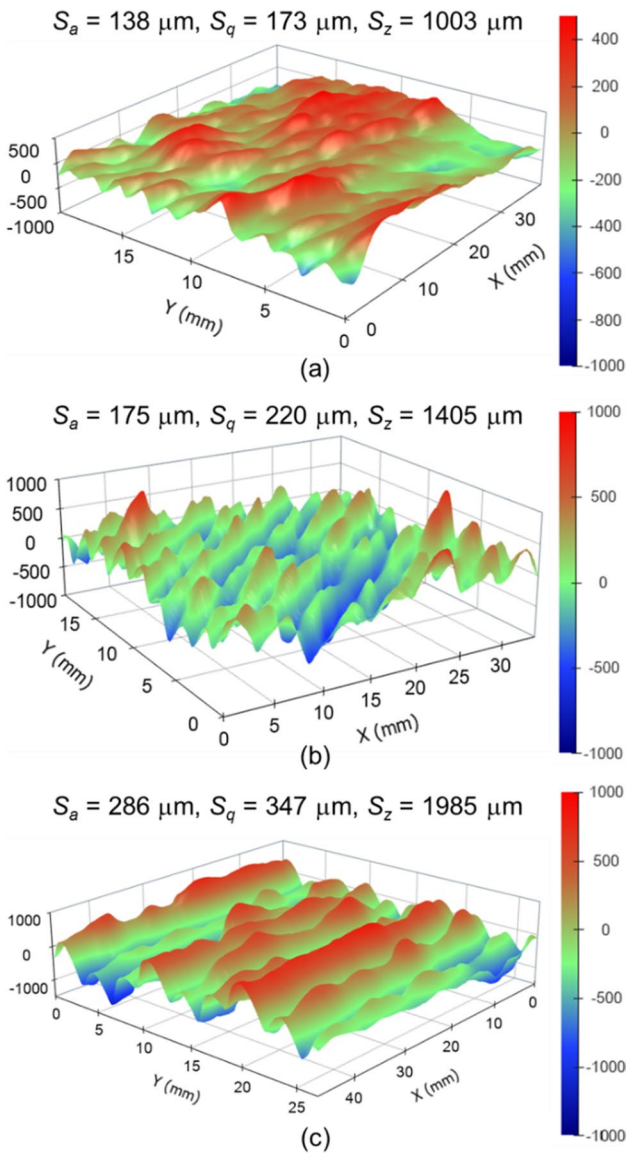


Fig. 4 **a** The surface morphology of the walls manufactured with 0.4, **b** 0.3, and **c** 0.2 m/min in deposition speed

increases, the heat input per length unit decreases, leading to a decrease in the layer thickness. As a result, the stair-stepping effect becomes weak, and the surface waviness mitigates. Conversely, as the travel speed decreases, the heat input per length unit increases, leading to an augmentation in the interlayer temperature, which extends the time of solidification and increases the fluidity of the melting pool. This also makes the surface significantly rougher [15, 40].

3.2 Microstructures

Figure 5a shows a typical microstructure characterization of a layer in the central part of the walls. All three wall reveals a similar trend of the microstructure evolution. The transition of microstructure between two adjacent layers is also visibly.

The microstructure of 308L stainless steel single-pass-multilayer walls primarily comprises two phases, namely, γ-austenite and δ-ferrite. The γ-austenite phases are dominant, while the δ-ferrite phases show a little quantity and exist in the austenite grains' boundaries. Based on the chemical composition analysis (Table 2), the development in microstructures of WAAMed stainless steel and the ferrite content can be predicted by using the pseudobinary phase schematic of the Cr-Ni–Fe system (Fig. 6b) and the Schaeffler's diagram (Fig. 6b), respectively. In this research, the equivalent chromium is calculated by $Cr_{eq}\% = (Cr + Mo + 1.5 \cdot Si + 0.5 \cdot Nb)\% = 20.83\%$, and the equivalent nickel is obtained by $Ni_{eq}\% = (Ni + 30 \cdot C + 0.5 \cdot Mn)\% = 11.20\%$, resulting in the ratio $\frac{Cr_{eq}}{Ni_{eq}} = 1.86$. This value falls in the range between 1.48 and 1.95. As a result, the percentage of ferrite is predicted at about 11–12% (Fig. 6a), and the solidification's mode of WAAMed stainless steel (308L) respects to the F/A (ferrite/austenite) mode (Fig. 6b): liquid → liquid + δ-ferrite → liquid + δ-ferrite + γ-austenite → δ-ferrite + γ-austenite [41].

Firstly, ferrite is precipitated from the liquid phase of melted metal because of the promotion of Cr, Si, Mo, and other elements. At the ending of primary ferrite solidification, the austenite phase is formed at boundaries of cellular grains or columnar dendrites of ferrites due to the peritectic-eutectic reaction. During the solidification time, δ-ferrite phases become increasingly unstable, and they are consumed by γ-austenite phases through the diffusion-controlled reaction [41]. Nevertheless, due to the fast cooling, there is little quantity of δ-ferrite phases that do not transform into γ-austenite completely, and they exist in γ-austenite

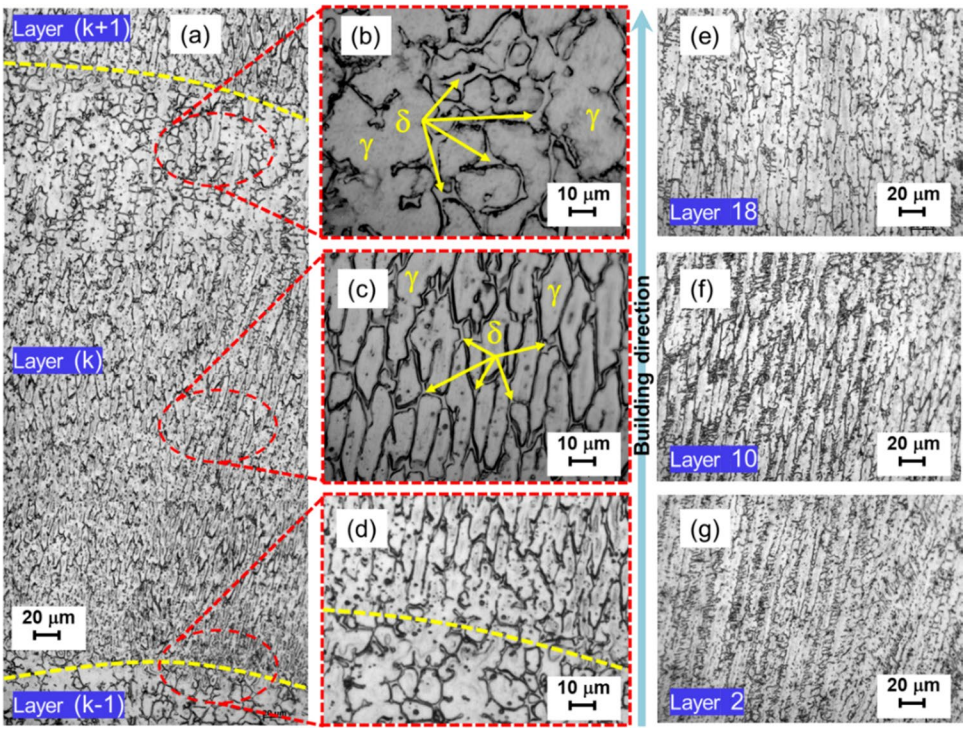


Fig. 5 **a** Typical microstructures of a layer in the walls' center, high magnification of microstructures at **b** the top, **c** at the middle, **d** at the interface of the layer, and **e**–**g** the microstructures at the middle of layers L18, L10, and L2 of the wall built with $v = 0.4$ m/min

Table 2 Chemical composition of WAAMed thin-wall components

Elements	Cr	Ni	Cu	Mo	Mn	Si	C	P	S	Fe
%wt	19.98	10.13	0.102	0.028	1.65	0.55	0.008	0.028	0.015	Bal

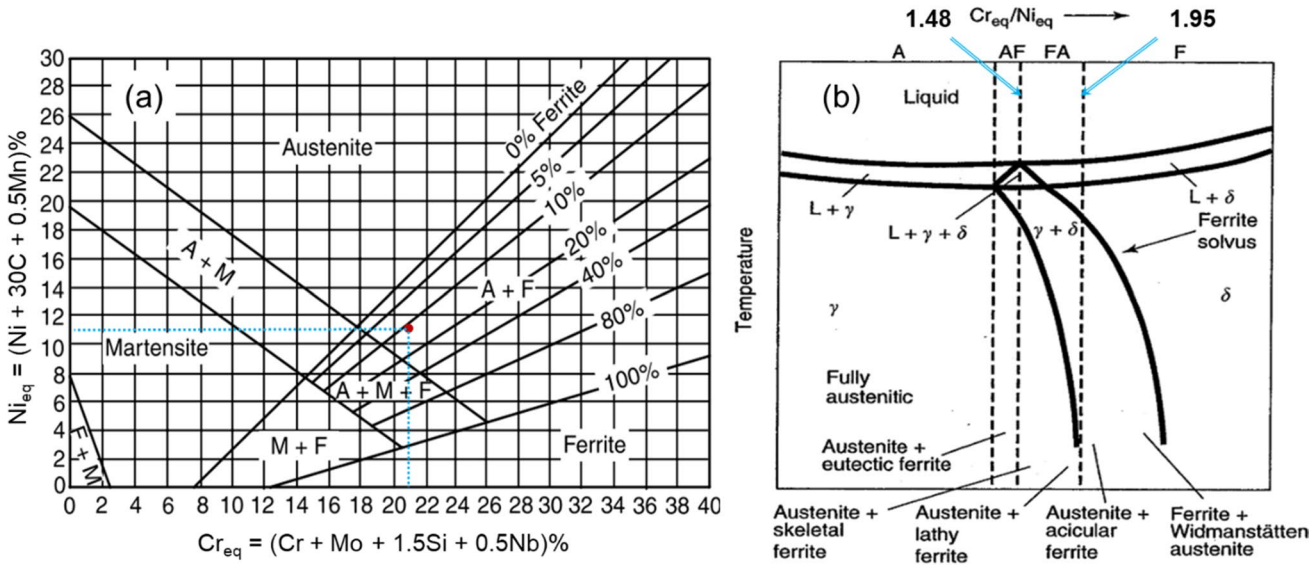


Fig. 6 **a** Schaeffler's diagram for estimating the ferrite content and **b** the pseudobinary phase schematic for predicting the solidification mode of stainless steels [41, 42]

boundaries at room temperature [43, 44]. The morphologies of residual δ-ferrite can be vermicular/skeletal (Fig. 5b, c, d) and lathy (Fig. 5f) shapes, depending on the cooling rates.

In a deposited layer (e.g., the layer k) in the central part of the thin-wall samples, the grains of γ-austenitic phases epitaxially grow from the interface with the layer $(k-1)$ and they develop with the cellular mode along the building direction (Fig. 5a, b, c, d). Further away from the bottom region, the cellular grains change to columnar grains (Fig. 5a, b, c). The columnar grains progressively enlarge to columnar dendrites and lastly transformed into equiaxed dendrites in the top part under the effect of heat source that forms the layer $(k+1)$ (Fig. 5a, b). This evolution trend of microstructures was also observed in previously published works [30, 43–45].

Moreover, as shown in Fig. 5e, f, g, which show the microstructure in the middle part of the layers L2, L10, and L18, respectively, the secondary-dendrite-arm spacing (SDAS) in the central of the layers enlarges along the levels of deposited layers. The average SDAS values have been measured to be 6.32 ± 0.97 µm (for L2; Fig. 5 g), 8.70 ± 1.02 µm (for L10; Fig. 5f), and 10.30 ± 1.38 µm for (L18; Fig. 5e). It is well known that the microstructure of the WAAMed is significantly affected by the thermal cycles [10, 46, 47]. Within the build of the wall, the dissipation of heat at different positions is very different. In the layers close to the substrate, the heat dissipates rapidly. The cooling rate becomes slower as the layers' level increases due to the heat accumulation phenomenon.

As revealed in Fig. 7, the temperature at the first peak and the second peak increases in order at points A, B, and C (1938 °C < 2060 °C < 2119 °C for the first peak and 1238 °C < 1434 °C < 1480 °C for the second peak of the thermal cycle curves). Similarly, the temperature at the first bottom valley (can be considered as the interlayer temperature during the WAAM process) increases progressively, whereas the average cooling rate of both the first and second cycles of the thermal cycle curve gradually decreases with an increase in the number of deposited layers. The average cooling rates of the first and second cycles at point A reveals the highest value (83 °C/s and 34 °C/s), followed by those at point B (57 °C/s and 23 °C/s) and at point C (53 °C/s and 20 °C/s), respectively. These results confirm that the heat accumulation occurs as the number of deposited layers increases, leading to an increase in the interlayer temperature increases and a reduction in the cooling rate. As the cooling rate decreases, the SDAS increases, as depicted by Eq. (3) [31, 48, 49]:

$$\mathrm{SDAS} = 50(\mathrm{CR})^{-0.4} \tag{3}$$

where CR is the cooling rate in K s^{-1} and SDAS in micrometers.

To analyze the effect of the deposition speed on the microstructure in terms of SDAS, the microstructures of the tenth layer (L10) of three walls were considered. Figure 8 shows the microstructure in the top (a, c, f), in the middle (b, d, g), and at the bottom (c, e, h) of the layer for the case $v = 0.4$, 0.3, and 0.2 m/min, respectively. It is revealed that, in all regions of the layer, the average SDAS increases as the deposition speed decreases. For instance, the average SDAS in the bottom of the layer is equal to 8.70 ± 1.02 µm,

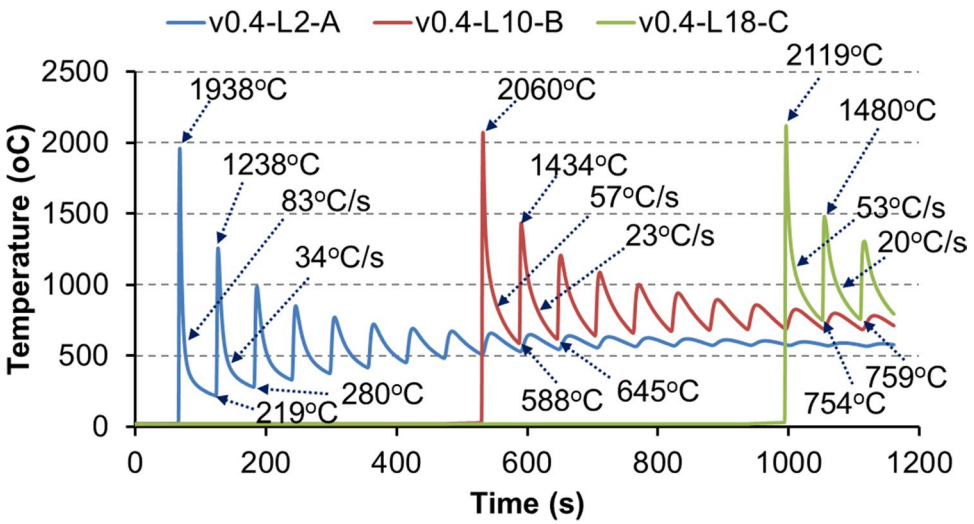


Fig. 7 Thermal cycles at the point A in L2, at the point B in L10, and at the point C in L18 of the wall built with $v = 0.4$ m/min

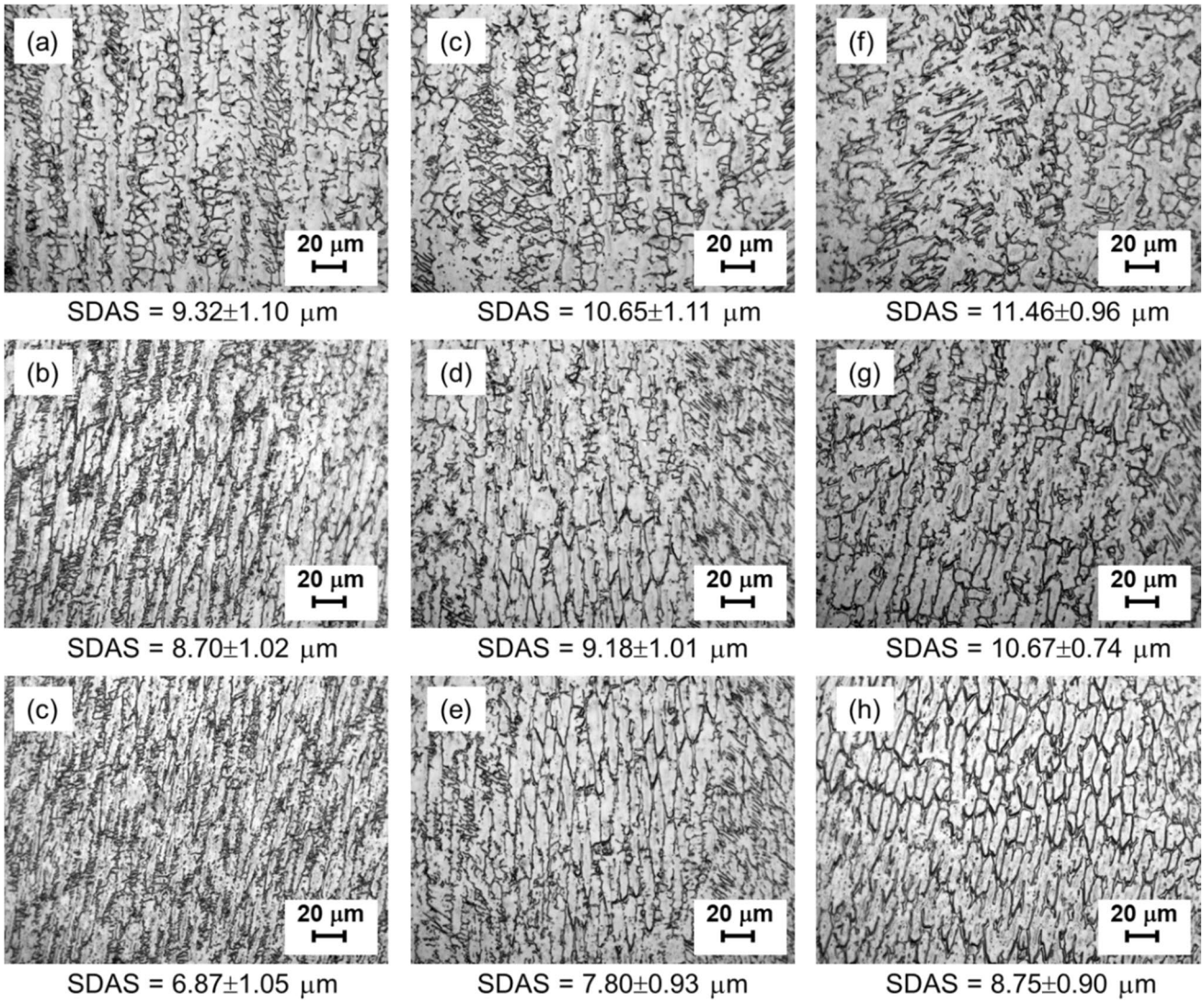


Fig. 8 The microstructure in the top (**a**, **c**, **f**), in the center (**b**, **d**, **g**) and in the bottom (**c**, **e**, **h**) of a specified layer in the central part of the walls produced with $v = 0.4$ m/min (**a**–**c**), 0.3 m/min (**c**–**e**), and 0.2 m/min (**f**–**h**)

9.18 ± 1.01 µm, and 10.67 ± 0.74 µm for the cases $v = 0.4$, 0.3, and 0.2 m/min, respectively.

These results in terms of microstructures were also consistent with the thermal evolution of the walls (Fig. 9). It is indicated that both the peak temperature and the interlayer temperature increase with a decrease in the deposition speed. The temperature at the first peak of the thermal cycle is about 2070 °C, 2248 °C, and 2453 °C, and the interlayer temperature is about 566 °C, 706 °C, and 830 °C for the cases $v = 0.4$, 0.3, and 0.2 m/min, respectively. Consequently, the cooling rate decreases as the deposition speed decreases. The average cooling rate value in the first thermal cycle is estimated to be about 57 °C/s, 46 °C/s, and 37 °C/s. Thereby, according to formula (3), the average SDAS augments with the decrease in the deposition speed.

Previous studies also demonstrated that the variation in the process parameters, resulting in the variation in the heat input causes changes in the heat accumulation and cooling rate in the walls [45, 48]. With higher levels of heat input applied, the cooling rate becomes lower, and resulting in larger SDAS. This trend is again confirmed by the results achieved in the current study.

3.3 Microhardness

Figure 10 reveals the microhardness (HV0.1) results measured in four layers (L8 to L11) in the middle part of the walls. In each layer, the measurement was conducted at nine points — three points at the bottom (B), three points at the middle (M), and three points at the top (T) of the layer. The measuring points of microhardness locate on the cross section's centerline. Figure 10a shows the comparative average microhardness in each region (B, M, and T) of four layers, while Fig. 10b compares the average microhardness of four layers between three walls.

It is firstly found that the average microhardness reduces along the bottom-to-top direction of a layer, for all layers. Indeed, the bottom of the layer features the finest columnar dendrites, which progressively enlarge towards the top of the layer. Finally, coarse equiaxed dendrites forms in the top of the layer (as illustrated in Fig. 8). Moreover, the microhardness in each region of a layer and the average microhardness in each layer decreases with the decrease in the deposition speed (Fig. 10a). Finally, the average microhardness of the four layers in the middle part of the walls decreases as the deposition speed decreases (Fig. 10b). The average microhardness in the middle of the wall fabricated with $v = 0.4$ m/min exhibits the highest values (164 ± 8.96 HV0.1), followed by $v = 0.3$ m/min (159 ± 6.32 HV0.1) and $v = 0.2$ m/min (153 ± 7.16 HV0.1), respectively. This result is consistent with the comparison of SDAS between three cases, as discussed in Sect. 3.3, because narrower SDAS in microstructures results in higher microhardness values [50].

3.4 Tensile properties

Table 3 and Fig. 11 summarize the tensile testing results of all specimens. Figure 11a shows typical engineering-stress–strain curves of three specimens {TS-B in the bottom, TS-M in the middle, and TS-T in the top} of a wall. These curves of WAAMed 308L stainless steel are composed of an elastic-deformation proportion and a plastic-deformation part before the specimens were broken. It is observed that the tensile characteristics of WAAMed specimens are influenced by their location in the walls. As the distance from the substrate increases, the tensile strengths (i.e., $\sigma^{(UTS)}$

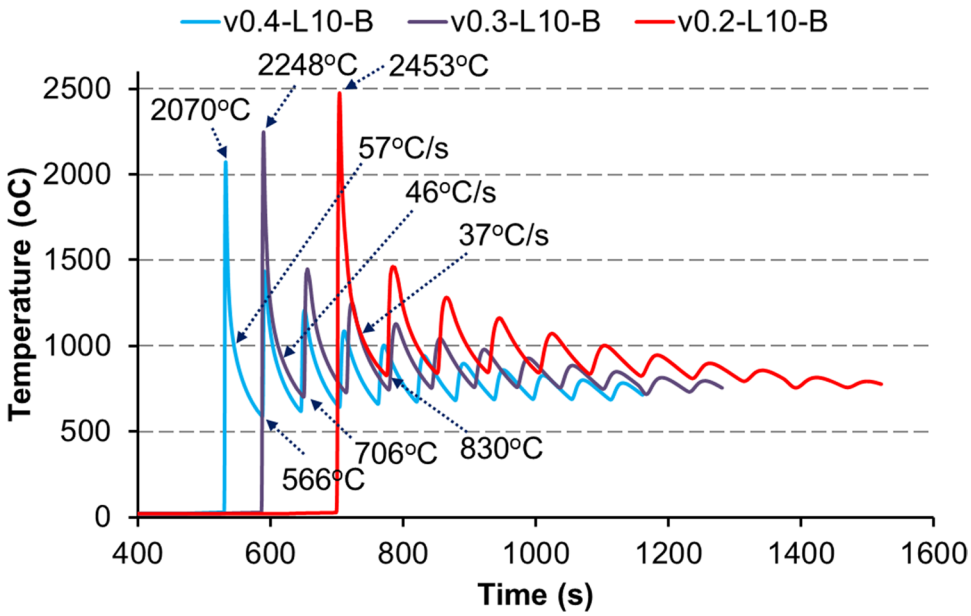


Fig. 9 The comparison of the thermal cycles at the point B in the layer L10 between the walls

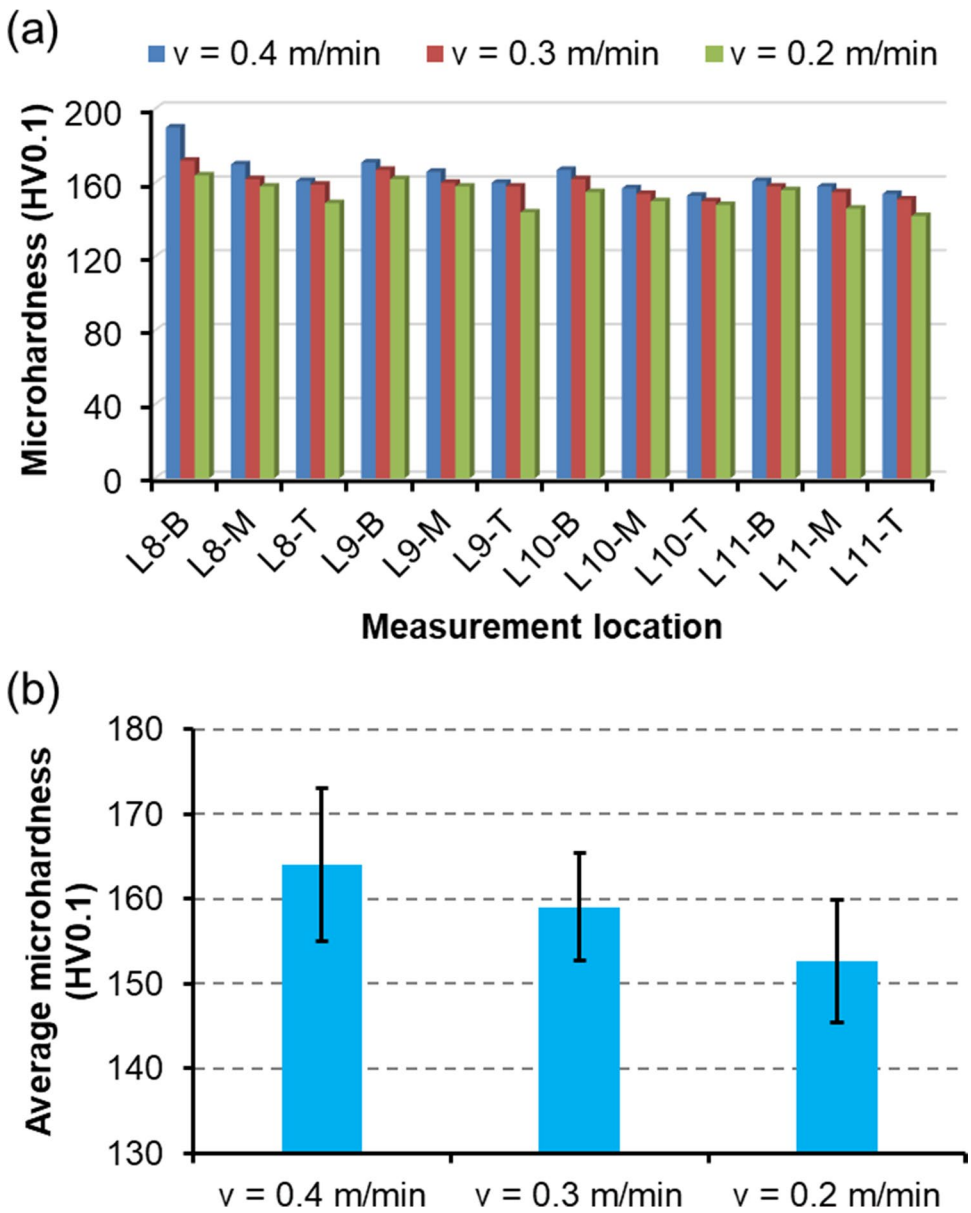


Fig. 10 **a** Microhardness in the bottom (B), in the middle (M), and in the top (T) of the layers (L8 to L11) of the walls, and **b** the comparison of global average microhardness in the middle part between the walls

Table 3 Tensile testing results

Specimens	$\sigma^{(YS)}$ (MPa)	$\sigma^{(UTS)}$ (MPa)	EL (%)
TS-T ($v=0.4$)	290	520	40
TS-M ($v=0.4$)	287	517	41
TS-B ($v=0.4$)	283	516	44
Average	**287 ± 3.51**	**518 ± 2.08**	**42 ± 2.08**
TS-T ($v=0.3$)	283	510	41
TS-M ($v=0.3$)	280	509	44
TS-B ($v=0.3$)	278	505	46
Average	**279 ± 2.52**	**507 ± 2.65**	**44 ± 2.52**
TS-T ($v=0.2$)	277	486	47
TS-M ($v=0.2$)	273	482	46
TS-B ($v=0.2$)	270	480	49
Average	**273 ± 3.51**	**483 ± 3.06**	**47 ± 1.53**

and $\sigma^{(YS)}$, MPa) of the horizontal tensile decrease: $\{\sigma^{(UTS)}, \sigma^{(YS)}\}_{(TS\text{-}T)} < \{\sigma^{(UTS)}, \sigma^{(YS)}\}_{(TS\text{-}M)} < \{\sigma^{(UTS)}, \sigma^{(YS)}\}_{(TS\text{-}B)}$. On the other hand, the total elongation (EL, %) generally increases with the increase of the distance from the substrate. For instance, in the case of the wall built with $v=0.4$ m/min, the values of $\sigma^{(UTS)}$ and $\sigma^{(YS)}$, and EL of the TS-T specimen are 520 MPa, 290 MPa, and 40%, respectively, followed by specimen TS-M ($\sigma^{(UTS)}=517$ MPa, $\sigma^{(YS)}=287$ MPa, EL = 41%), and specimen TS-B ($\sigma^{(UTS)}=516$ MPa, $\sigma^{(YS)}=283$ MPa, EL = 44%), respectively.

This anisotropy in mechanical properties is due to the inhomogeneity in microstructures of the as-built walls. Herein, the microstructures varied region by region in each layer and layer by layer along the building direction. Therefore, the mechanical properties of the wall material were

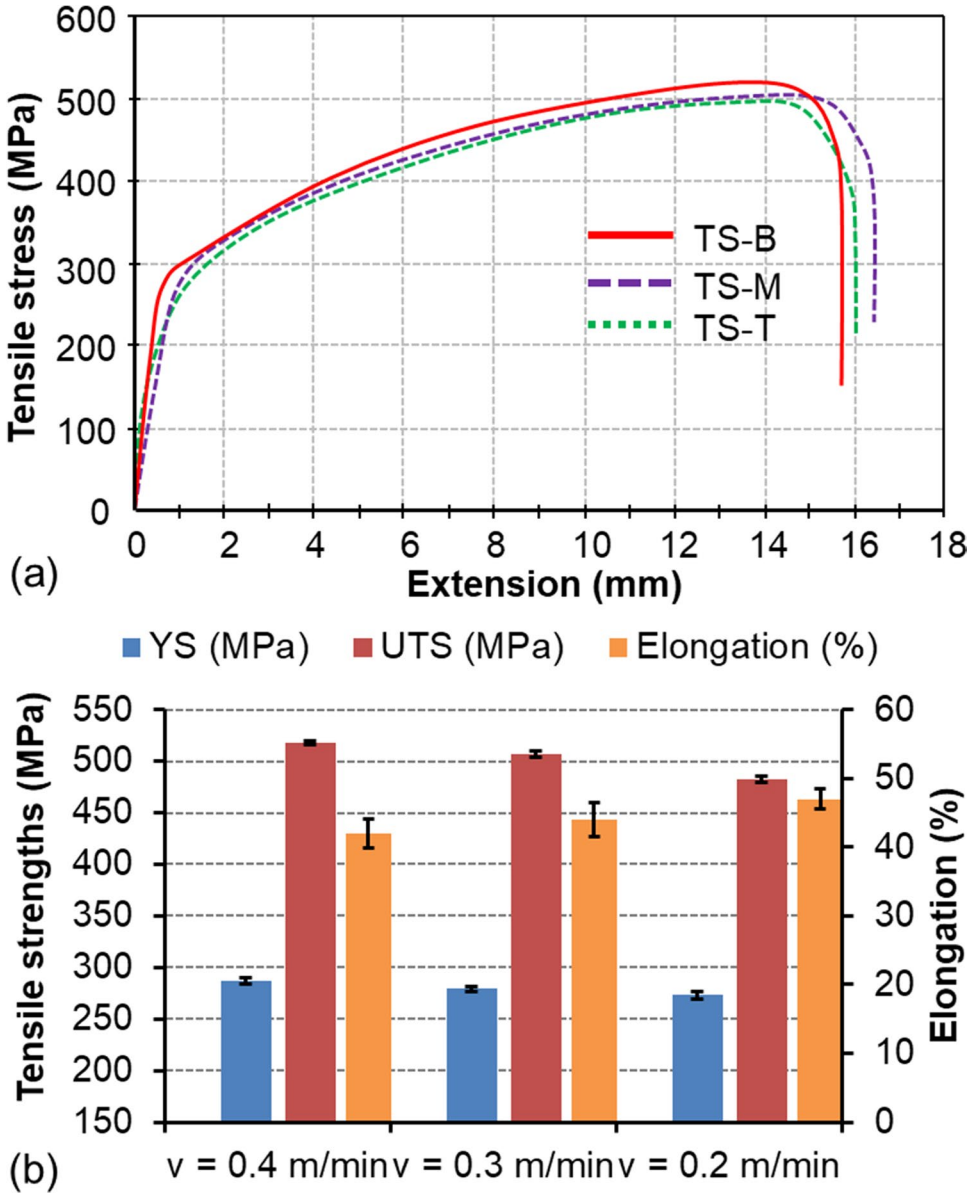


Fig. 11 **a** The typical stress–strain curves of the tensile specimens in different regions and **b** the comparison of tensile properties between the walls

anisotropic. Moreover, the variation of tensile strengths of WAAMed walls is related to the size of austenite grains in microstructures and the phase fraction [51, 52]. The relationship between the grain size and the yield strength of metallic materials is depicted by Hall–Petch formula [53]:

$$\sigma^{(YS)} = \sigma^{(0)} + \frac{k^{(YS)}}{\sqrt{d^{(G)}}} \tag{4}$$

where $\sigma^{(YS)}$ presents the yield strength (YS), $\sigma^{(0)}$ and $k^{(YS)}$ are the constant characteristics of metallic materials, and $d^{(G)}$ depicts the size of grains. According to this relationship, tensile properties become better as smaller SDAS in microstructures was produced. As a result, the tensile strengths ($\sigma^{(UTS)}$, $\sigma^{(YS)}$) decrease from the bottom to the top part of the walls. On the other hand, the total elongation and toughness of metallic alloys generally show an opposite trend. An increase in the grain size results in an augmentation in toughness and total elongation [51, 52].

Figure 11b shows the comparison on the average tensile strength and elongation between three cases. It is indicated that the decrease in the deposition speed (from 0.4 to 0.2 m/min) leads to a slight reduction in tensile strengths. The average value of $\sigma^{(UTS)}$ and $\sigma^{(YS)}$ in each case are as follows: $\{\sigma^{(UTS)} = 518 \pm 2.08$ MPa, $\sigma^{(YS)} = 287 \pm 3.51$ MPa for $v = 0.4$ m/min$\}$, $\{\sigma^{(UTS)} = 507 \pm 2.65$ MPa, $\sigma^{(YS)} = 279 \pm 2.52$ MPa for $v = 0.3$ m/min$\}$, and $\{\sigma^{(UTS)} = 483 \pm 3.06$ MPa, $\sigma^{(YS)} = 273 \pm 3.51$ MPa for $v = 0.2$ m/min$\}$ (Table 3). Meanwhile, the total elongation increases with the decrease in the deposition speed.

The wall fabricated with $v = 0.2$ m/min shows the highest average value of total elongation ($EL = 47 \pm 1.53\%$), followed by the walls produced with $v = 0.3$ m/min ($EL = 44 \pm 2.52\%$) and 0.4 m/min ($EL = 42 \pm 2.08\%$), respectively.

The main reason is the SDAS increases with the decrease in the deposition speed, as discussed in Sect. 3.3. The refinement in microstructures significantly influences the strengths of metallic materials. According to the Hall–Petch relationship, as shown in Eq. (4), the tensile strengths become better as smaller SDAS in microstructures was produced. As a result, the walls fabricated with a higher deposition speed have better tensile characteristics.

Moreover, the tensile strengths of the as-built material (308L) can be comparable or higher than those of annealed/rolled 308L stainless steel. As aforementioned, the tensile strengths of as-fabricated 308L steel walls are about 483–518 MPa in UTS and 273–287 MPa in YS (Table 3), whereas the values of UTS and YS of annealed/rolled 308L stainless steel are about 515 and 205 MPa, respectively [54].

Figure 12 shows the fractured surfaces' morphology of tensile testing specimens (TS-Mi) in all three cases. The fractured surfaces present many dimples in equiaxed shapes, indicating the ductile fracturing mode of 308L stainless steel produced by WAAM, and it features a great toughness [55, 56]. It is also revealed that the wall built with the lowest deposition speed ($v = 0.2$ m/min) exhibits larger and deeper dimples (Fig. 12c). On the other hand, in the case of the wall produced with $v = 0.4$ m/min, the dimples are shallow and have the smallest size (Fig. 12a). The size and depth of dimples are normally associated with the grain size, and these parameters increase as the grain size increases [52]. As previously discussed, the wall built with $v = 0.4$ m/min shows the smallest grain size, followed by the walls deposited with $v = 0.3$ and 0.2 m/min, respectively (Fig. 8). Thereby, the dimples reveal the smallest size in the case of

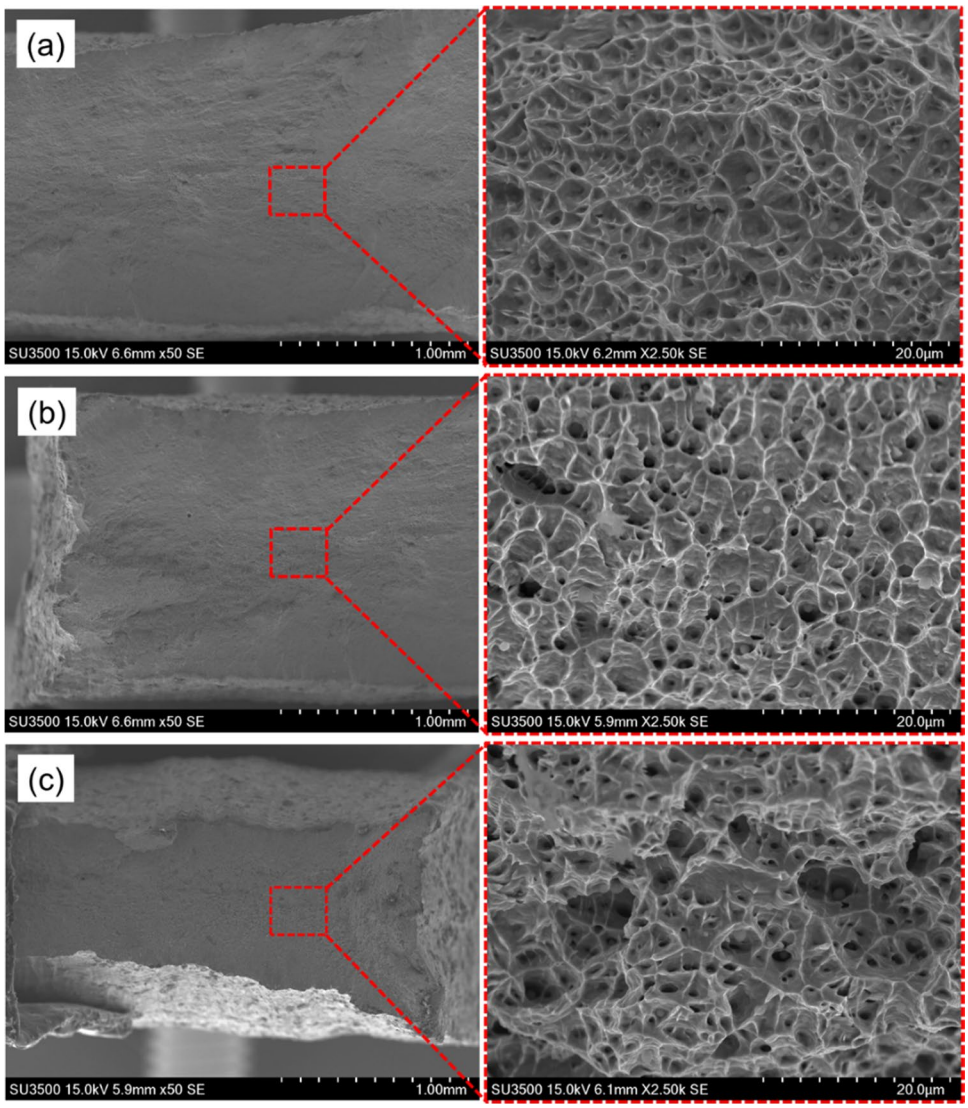


Fig. 12 Fractured surface morphologies of the tensile samples produced with **a** $v = 0.4$ m/min, **b** $v = 0.3$ m/min, and **c** $v = 0.2$ m/min

$v = 0.4$ m/min (Fig. 12a). Deeper dimples appearing in the fractured surface indicate the specimen having higher ductility. Herein, the depth of dimples increases when the travel speed increases. As a result, the sample built with $v = 0.2$ m/min shows the highest elongation. This again confirms the walls fabricated with lower deposition speeds having better toughness.

4 Conclusions

This study gives a deep insight into the influence of thermal cycles and the deposition speed in WAAM on the geometrical and metallurgical characteristics of 308L stainless steel. Three levels of the deposition speed were utilized to produce single-pass multi-layer walls. The shape, surface morphology, microstructures, and mechanical properties of the walls have been examined. The major outcomes of the work are depicted as follows:

- The walls produced with higher deposition speed feature smoother side surfaces. However, they are lower in height and layer thickness, as well as thinner in width. The surface waviness (S_a) of the thin walls reveals a magnitude from 138 to 286 µm. Therefore, the as-built surfaces generally required finishing machining operations to meet technique specifications of functional ones.
- The microstructure evolution of the walls produced with different deposition speed exhibits a similar trend. The microstructures mainly consist of primarily γ-austenite phases and little quantity of residual δ-ferrite phases in lathy and skeletal or vermicular shapes. The microstructures are the results of complex thermal cycles, which occur during the deposition. In a layer, the columnar grains are formed in the bottom; they gradually enlarge towards the top, and finally transformed into equiaxed dendrites in the top zone of the layer.
- With an increase in the deposition speed, the secondary dendrite arms' spacing (SDAS) becomes narrow, thus enhancing the microhardness and tensile strengths. The microhardness of 308L stainless steel walls produced by WAAM falls in a range from 153 ± 7.16 HV0.1 to 164 ± 8.96 HV0.1, and the tensile strengths of the as-fabricated material can be comparable or higher than those of annealed/rolled 308L stainless steel — e.g., 483–518 MPa vs. 515 MPa in UTS and 273–287 MPa vs. 205 MPa in YS, respectively.

Author contribution Van Thao Le: conceptualization, methodology, data analysis, writing — original draft preparation, reviewing and editing; Dinh Si Mai, Duc Manh Dinh, Van Anh Nguyen: validation, investigation; Manh Cuong Bui, Duc Manh Dinh, Van Canh Nguyen: formal analysis, software, data curation; Kilian Wasmer, Van Anh Nguyen, Duong Vu: reviewing, and editing.

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Data availability All data generated or analyzed during this study are included in this published article.

Declarations

Conflict of interests The authors declare no competing interests.

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